

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459471.8825 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.042
Transit depth (Rp/Rs)^2	1.69 +/- 1.09 [%]
Orbital Inclination (inc)	89.14 +/- 3.0
Airmass coefficient 1 (a1)	1.012 +/- 0.019
Airmass coefficient 2 (a2)	-0.0014 +/- 0.0061
Scatter in the residuals of the lightcurve fit is	1.87 %
Best Comparison Star	#2 - [401, 226]
Optimal Aperture	4.17
Optimal Annulus	10.44
Transit Duration (day)	0.105 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.13 ± 0.042 vs 0.1618 (deviation 0.52σ)
Duration fit vs published	0.105 d vs 0.0721 d (deviation 0.97σ)
Mid-time O–C	18.8 min
Depth SNR	2.1
Residual scatter	1.87 %

Light curve

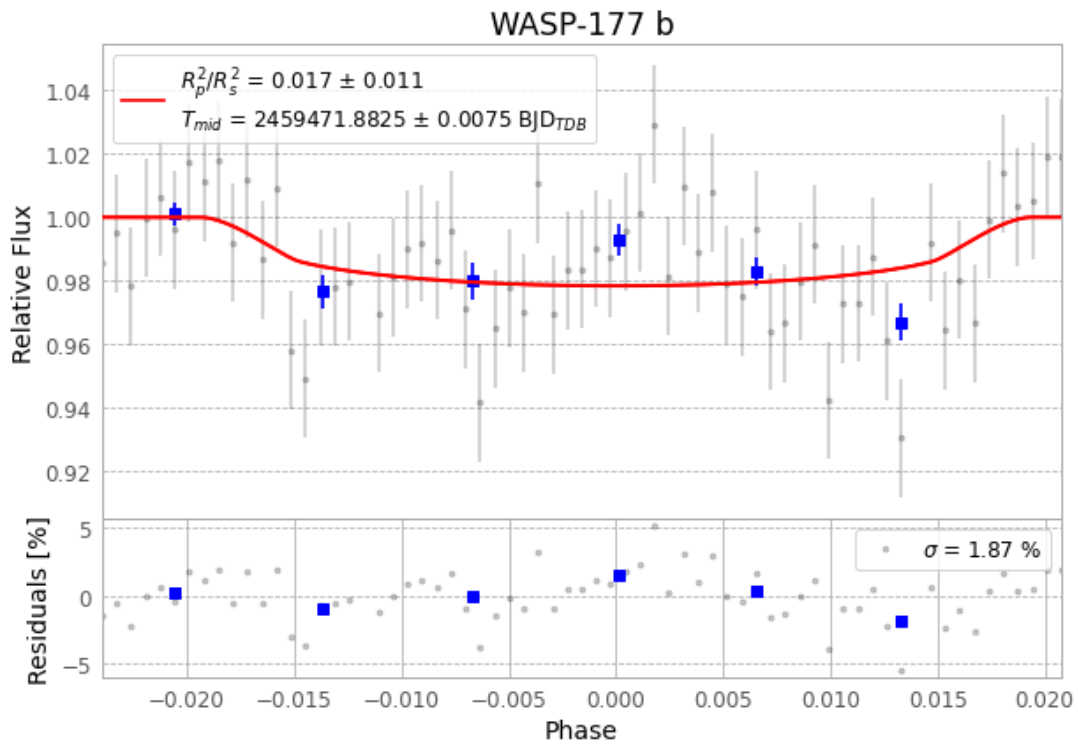


Figure 1 — FinalLightCurve_WASP-177 b_2021-09-14.png (SHA-256 742284b1c8278557...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 111], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 77e8f28970b0b188...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), WASP-177210914071512.FITS → WASP-177210914103312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-210914.log (0c865b002ecd...), FinalLightCurve_WASP-177 b_2021-09-14.png (742284b1c827...), FinalLightCurve_WASP-177 b_2021-09-14.csv (0c0e698fb736...)

Compute resources

Wall time 112.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:45Z.